

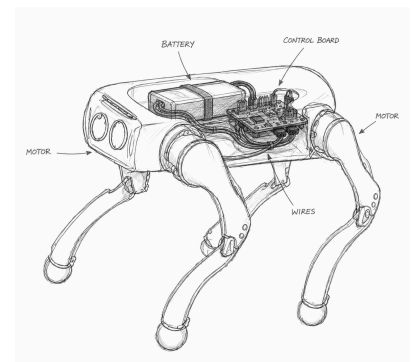
First mechanical assignment

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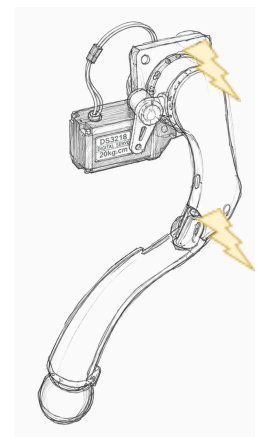
1. Robot Body and Chassis

The robot has a rectangular PLA chassis with rounded corners that supports most of the main components, such as the battery and motors. Each of the four legs is attached to one corner of the chassis, for walking and balance.



2. Leg Design

Each leg has two main links: the femur and the tibia. A rubber pad is attached to the end of the leg to serve as the foot, giving necessary friction during movement. The femur is connected to a dedicated servo motor, allowing independent movement of all four legs.



3. Degrees of Freedom

Each leg of the robot has two joints, resulting in 2 DOF per leg. Meaning the robot has a total of **8 DOF**.

4. Actuator Selection

The robot uses MG996R servo motors to actuate the leg joints.



5. Torque Calculation

Assuming the total mass of the robot is **1.15 kg**

- total weight: $F = m \cdot g$, $F = 1.15 \times 9.81 = \mathbf{11.28\text{ N}}$
- the load acting on one leg is **2.82N**
- Assuming the distance from the joint to the foot is **0.1 m**, the required torque is $\tau = F \times r$ $\tau = 2.82 \times 0.1 = \mathbf{0.282\text{ Nm}}$

6. Center of Gravity and Stability

The center of gravity is in the center of the chassis because most of the robot's heavy components, such as the battery and control board, are inside the main body. This provides balanced stability during movement.

7. Proposed Walking Method

The proposed walking method is the Trot Gait, in which diagonal pairs of legs move together. This gait offers a good balance between speed and stability, making it suitable for a lightweight quadruped robot.

8. Expected Mechanical Issues

- motor overload
- Joint backlash.
- Slipping on smooth surfaces.